# Introduction.

Current document describes full control and data exchange between controller (master) and heater (slave) using CAN bus. It's allowed to have up to 99 heaters connected to the same CAN bus, however the condition must be observed that there are no connected heaters with the same last two digits of the serial number.

# CAN bus specification.

Device supports **CAN 2.0B** version (29-bit identifier), data size is 8 bytes, communication speed 250Kbps.

# Description of CAN bus frames.

CAN bus frame ID is 29-bits long, it contains such elements like:

- Frame ID
- Direction of message (Master -> Slave or Slave -> Master)
- Command ID
- Slave device ID

Here is example of CAN bus frame – **0x1C05012E**

**1C** - First byte of frame ID always equals to 0x1C and allows to recognize purpose of frame.

**05** – This byte allows to recognize data direction, following options available:

**0x05** - Master sends data to Slave device
**0x07** - Slave device sends data to the Master device
**0x09** - Slave device sends data to Slave device (can be implemented in future)

**01** – Represents command number. Detailed description of all commands can be found later in this document.

**2E** – Last byte is ID number for Slave device. This allows to address data to specific Slave device in case if there are several Slave devices in CAN bus. Master device also can recognize which Slave device is sending data/response. If master device needs to address data to all Slave devices same time, ID number must be equal to **0xFF**. Detailed description about method how Slave device ID is created can be found later in this document.

# Heater commands.

Table below represents summary of all available commands:

| Command ID | Description |
|---|---|
| **0x01** | Start heating, Set work mode/heating parameters |
| **0x02** | Stop heating |
| **0x03** | Heater status info |
| **0x04** | Request heater's ID |
| **0x05** | Master device presence |

## Start heating — Command ID: 0x01

This command is used to start heating, define heating work mode and heating parameters.

| Number | Name | Type | Description |
|---|---|---|---|
| **Byte 0** | Work mode | uint8 | This byte defines heater work mode:<br><br>**0x00** – Will keep existing working mode<br>**0x01** – Temperature mode<br>**0x02** – Power mode<br>**0x03** – Ventilation mode |
| **Byte 1** | Target temperature | uint8 | This byte defines target temperature when heater operates in temperature mode. Available range is 0-30 degrees.<br><br>**0xFF** – Keep temperature unchanged<br>**0x00-0x1E** – Temperature value |
| **Byte 2** | Power level | uint8 | This byte defines constant power level when heater operates in Power mode. Available range is 1-6. Any value, which will be greater than 6 will set heating power equal to 6.<br><br>**0x00** – Keep existing power level<br>**0x01 – 0x06** – Heating power level |
| **Byte 3** | Reference temperature | uint8 | This byte defines source of reference temperature when heater operates in Temperature mode.<br><br>**0x00** – Keep temperature source unchanged<br>**0x01** – External temperature sensor |

| | | | |
|---|---|---|---|
| | | | **0x02** – Master device provides reference temperature |
| **Byte 4** | Heat duration H | uint16 | This byte represents duration (in minutes) of heating or ventilation.<br><br>**0x0000** – Keep existing duration<br>**0x0001 – 0xFFFE** – Duration of heating |
| **Byte 5** | Heat duration L | | |
| **Byte 6** | Not used | uint8 | Reserved for future purpose |
| **Byte 7** | Not used | | Reserved for future purpose |

# Stop heating Command ID: 0x02

This command is used to stop heater.

| Number | Name | Type | Description |
|---|---|---|---|
| **Byte 0** | Not used | | |
| **Byte 1** | Not used | | |
| **Byte 2** | Not used | | |
| **Byte 3** | Not used | | |
| **Byte 4** | Not used | | |
| **Byte 5** | Not used | | |
| **Byte 6** | Not used | | |
| **Byte 7** | Not used | | |

# Heater Status info Command ID: 0x03

This message is periodically (**once per 500ms**) generated by Slave device and sent over CAN bus. This message can be also used to determine how many Slave devices already connected to the CAN bus.

| Number | Name | Type | Description |
|---|---|---|---|
| **Byte 0** | Heater model | uint8 | **0x00** – Heater not detected<br>**0x01** – Air 2D (old model)<br>**0x02** – Air 2D<br>**0x03** – Air 8D<br>**0x04** – Air 9D<br>**0x05** – Air 4D<br>**0x06** – Flow 5<br>**0x07** – Flow 14 |
| **Byte 1** | Actual condition | uint8 | **0x00** – Heater in standby<br>**0x01** – Starting<br>**0x02** – Ignition<br>**0x03** – Burning<br>**0x04** - Purge |
| **Byte 2** | Error code | uint8 | **0x00** – No errors codes<br>Any other value represents actual error code |
| **Byte 3** | Working mode | uint8 | **0x00** – Unknown<br>**0x01** – Temperature mode<br>**0x02** – Power mode<br>**0x03** – Ventilation mode |
| **Byte 4** | Supply voltage H | uint16 | Value represents heater's supply voltage in mV |
| **Byte 5** | Supply voltage L | | |
| **Byte 6** | Temperature | int8 | For air heaters value represents air intake temperature of heater -128...+127 (in Celsius)<br>For Flow heaters temperature represents water temperature -128…+127(in Celsius) |
| **Byte 7** | Not used | uint8 | Reserved for future purpose |

# Request heater's ID Command ID: 0x04

This command is used to request serial number and total working hours from heater. Slave device will ignore contents of payload (Byte 0 – Byte 7) on request.

Response from Slave device:

| Number | Name | Type | Description |
|---|---|---|---|
| **Byte 0** | Total hours H | uint16 | Value represents total working hours |
| **Byte 1** | Total hours L | | |
| **Byte 2** | Num1 | uint24 | To get serial number, use following method (Num1*65536) + (Num2*256) + Num3 |
| **Byte 3** | Num2 | | |
| **Byte 4** | Num3 | | |
| **Byte 5** | Not used | | |
| **Byte 6** | Not used | | |
| **Byte 7** | Not used | | |

# Master device presence Command ID: 0x05

This message is mandatory and is used to inform Slave devices about presence of Master device. If during heating Mater will stop sending presence message, all Slave devices will stop heating and trigger error code. Master device must send presence message **once per 1second.** Current presence message is also used to inform Slave devices about room temperature, which is provided by Master device.

| Number | Name | Type | Description |
|---|---|---|---|
| **Byte 0** | Temperature | int8 | Current byte represents room temperature; range is -127…+127.<br>If room temperature not available, master must send value **0x80** |
| **Byte 1** | Not used | | |
| **Byte 2** | Not used | | |
| **Byte 3** | Not used | | |
| **Byte 4** | Not used | | |
| **Byte 5** | Not used | | |
| **Byte 6** | Not used | | |
| **Byte 7** | Not used | | |

# Slave device ID byte.

To address message to specific Slave device, last byte of CAN bus frame contains ID of Slave device. There are two ways to get Slave device ID:

1. Connect Slave device to the CAN bus and wait till **Heater Status Info** message is received from Slave device. Last byte of frame ID from status message will represent Slave device ID byte.
2. Slave ID byte can be calculated using heater's serial number, which can be found on heater. Picture below shows example of heater's serial number:




Using last two digits from serial number (in this example it is 48), convert decimal value to the hex, as result it will be Slave device ID number on CAN bus.

Examples:

| Last two digits of serial number | Value after conversion to hex |
|---|---|
| **48** | 0x30 |
| **00** | 0x00 |
| **01** | 0x01 |
| **99** | 0x63 |